

Numerical Methods in Engineering and Applied Science

Class Project 02

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Question: 01

If a debt is amortized by regular payments of size R and is subject to interest I , the unpaid balance is P_k , where $P_{k+1} = (1 + I)P_k - R$. The initial debt being $P_0 = A$. What is the unpaid balance as a function of A, I , and R . Also, what must be the regular payment R be in order to reduce the balance exactly to zero in N payments?

Solution:

The recurrence relation governing the unpaid balance is:

$$P_{k+1} = (1 + I)P_k - R$$

Where:

- P_k is the unpaid balance after k payments,
- I is the interest rate per period,
- R is the fixed regular payment,
- A is the initial debt, i.e., $P_0 = A$.

We aim to find a closed-form expression for P_k .

Compute Unpaid Balances for Early Periods

Let's compute the unpaid balance step by step for the first few periods to identify a pattern.

For $k = 0$:

$$P_0 = A$$

For $k = 1$:

$$P_1 = (1 + I)P_0 - R = (1 + I)A - R$$

For $k = 2$:

$$P_2 = (1 + I)P_1 - R = (1 + I)((1 + I)A - R) - R$$

Simplify:

$$P_2 = (1 + I)^2 A - (1 + I)R - R$$

For $k = 3$:

$$P_3 = (1 + I)P_2 - R = (1 + I)^3 A - (1 + I)^2 R - (1 + I)R - R$$

General Formula for P_k

From the pattern, the unpaid balance after k payments is:

$$P_k = (1 + I)^k A - R((1 + I)^{k-1} + (1 + I)^{k-2} + \cdots + 1)$$

The summation $((1 + I)^{k-1} + (1 + I)^{k-2} + \cdots + 1)$ is a finite geometric series, which simplifies to:

$$\text{Sum} = \frac{(1 + I)^k - 1}{I}$$

Thus, the general formula for P_k is:

$$P_k = (1 + I)^k A - R \cdot \frac{(1 + I)^k - 1}{I}$$

Solve for R

To find the regular payment R , we require that the balance is exactly zero after N payments:

$$P_N = (1 + I)^N A - R \cdot \frac{(1 + I)^N - 1}{I} = 0$$

Rearranging for R :

$$R = \frac{(1 + I)^N A}{\frac{(1 + I)^N - 1}{I}}$$

Simplify:

$$R = A \cdot \frac{I(1 + I)^N}{(1 + I)^N - 1}$$

Question: 02

The differential equation describing the ice build-up on a lake is given by:

$$h^2 \frac{d^2 h}{dt^2} + 2 \left(\frac{dh}{dt} \right)^2 + 4L \left(\frac{1+h}{2+h} \right) \frac{dh}{dt} - \frac{4L}{2+h} = 0, \quad (0.0.1)$$

Where h is the thermal resistance of the ice slab (thickness divided by thermal conductivity), L is a dimensionless latent heat for which 0.5 is a reasonable value, and t is a non-dimensional time. Appropriate initial conditions are $h(0) = 0.01$, $h'(0) = 0.5$. Solve the problem for all values of t until $h = 2$.

Solution:

The second-order differential equation can be rewritten as a system of two first-order differential equations:

$$y_1 = h, \quad y_2 = \frac{dh}{dt}.$$

Thus, the system becomes:

$$\begin{aligned} \frac{dy_1}{dt} &= y_2, \\ \frac{dy_2}{dt} &= \frac{-2y_2^2 - 4L \left(\frac{1+y_1}{2+y_1} \right) y_2 + \frac{4L}{2+y_1}}{y_1^2}. \end{aligned}$$

We solve this system numerically using the fourth-order Runge-Kutta (RK4) method. The RK4 method proceeds as follows:

1. Define step size h and initialize t_0 , $y_1(0)$, and $y_2(0)$.
2. Compute intermediate slopes:

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right), \\ k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right), \\ k_4 &= f(t_n + h, y_n + k_3). \end{aligned}$$

3. Formula:

$$y_{n+1} = y_n + \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}.$$

4. Repeat until the stopping condition $y_1 = 2$ is met.

Results

Using numerical integration, the solution for $h(t)$ is computed until $h = 2$. The computation involves determining the thermal resistance of the ice slab as a function of time. The results are shown in the figure below.

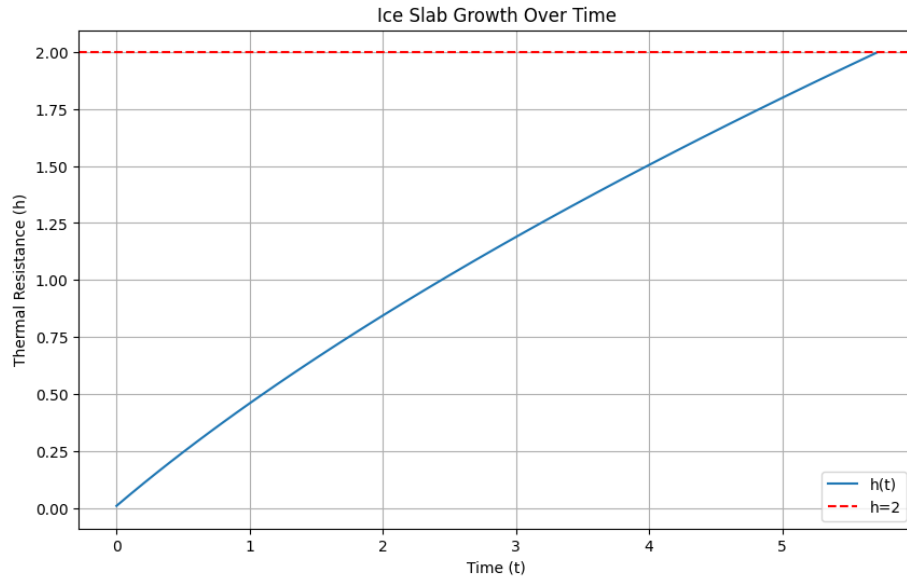


Figure 1: Thermal resistance $h(t)$ as a function of non-dimensional time t .

The solution shows that $h = 2$ is reached at approximately $t \approx 5.72$.

Question: 03

Find the multistep method of the form:

$$y_{n+2} + \alpha_1 y_{n+1} + \alpha_0 y_n = h(\beta_1 f_{n+1} + \beta_0 f_n),$$

of the highest possible order. Try this formula on the example $y' = y, y(0) = 1$
 $h = 1/10, 1/20, 1/40$. Explain the results.

Solution:

Expand y_{n+2} , y_{n+1} , and y_n Using Taylor Series

Expand y_{n+2} and y_{n+1} about t_n :

- For y_{n+2} :

$$y_{n+2} = y_n + 2hy'_n + 2h^2y''_n + \frac{4h^3}{6}y'''_n + \mathcal{O}(h^4).$$

Since $y' = y$, $y'' = y$, and $y''' = y$, substitute:

$$y_{n+2} = y_n + 2hy_n + 2h^2y_n + \frac{4h^3}{6}y_n + \mathcal{O}(h^4).$$

Simplify:

$$y_{n+2} = y_n(1 + 2h + 2h^2 + \frac{4h^3}{6}) + \mathcal{O}(h^4).$$

- For y_{n+1} :

$$y_{n+1} = y_n + hy'_n + \frac{h^2}{2}y''_n + \frac{h^3}{6}y'''_n + \mathcal{O}(h^4).$$

Substitute derivatives:

$$y_{n+1} = y_n + hy_n + \frac{h^2}{2}y_n + \frac{h^3}{6}y_n + \mathcal{O}(h^4).$$

Simplify:

$$y_{n+1} = y_n(1 + h + \frac{h^2}{2} + \frac{h^3}{6}) + \mathcal{O}(h^4).$$

Substitute Into the Method

Substitute the Taylor expansions of y_{n+2} and y_{n+1} into the formula:

$$y_{n+2} + \alpha_1 y_{n+1} + \alpha_0 y_n = h(\beta_1 y_{n+1} + \beta_0 y_n).$$

Expand each term:

- Substitute y_{n+2} :

$$y_{n+2} = y_n(1 + 2h + 2h^2 + \frac{4h^3}{6}).$$

- Substitute y_{n+1} :

$$y_{n+1} = y_n(1 + h + \frac{h^2}{2} + \frac{h^3}{6}).$$

Combine all terms into the method:

$$y_n(1 + 2h + 2h^2 + \frac{4h^3}{6}) + \alpha_1 y_n(1 + h + \frac{h^2}{2} + \frac{h^3}{6}) + \alpha_0 y_n = h(\beta_1 y_n(1 + h + \frac{h^2}{2} + \frac{h^3}{6}) + \beta_0 y_n).$$

Comparing Coefficients for h^0, h^1, h^2, h^3

Group terms by powers of h and set up equations for the coefficients:

- h^0 :

$$1 + \alpha_1 + \alpha_0 = 0.$$

- h^1 :

$$2 + \alpha_1 = \beta_1 + \beta_0.$$

- h^2 :

$$2 + \frac{\alpha_1}{2} = \beta_1.$$

- h^3 :

$$\frac{4}{3} + \frac{\alpha_1}{6} = \frac{\beta_1}{2}.$$

Solve the System of Equations

Using these equations:

- From **Second-Order**:

$$2 + \frac{\alpha_1}{2} = \beta_1 \implies \beta_1 = 2 + \frac{\alpha_1}{2}.$$

- From **Third-Order**:

$$\frac{4}{3} + \frac{\alpha_1}{6} = \frac{\beta_1}{2}.$$

Substitute $\beta_1 = 2 + \frac{\alpha_1}{2}$:

$$\frac{4}{3} + \frac{\alpha_1}{6} = \frac{1}{2}\left(2 + \frac{\alpha_1}{2}\right).$$

Solve for α_1 :

$$\alpha_1 = 4.$$

- From **Zeroth-Order**:

$$1 + \alpha_1 + \alpha_0 = 0 \implies \alpha_0 = -5.$$

- From **First-Order**:

$$2 + \alpha_1 = \beta_1 + \beta_0 \implies \beta_0 = 2.$$

Final coefficients:

$$\alpha_1 = 4, \quad \alpha_0 = -5, \quad \beta_1 = 4, \quad \beta_0 = 2.$$

Final Multistep Method

The method becomes:

$$y_{n+2} + 4y_{n+1} - 5y_n = h(4y_{n+1} + 2y_n).$$

h	t	y_{approx}	y_{exact}	Error
1/10	0	1.00000	1.00000	0.00000
	1/10	1.10000	1.10517	0.00517
	1/5	1.24000	1.22140	0.01860
	3/10	1.25600	1.34986	0.09386
	2/5	1.92640	1.49183	0.43457
	1/2	-0.40384	1.64872	2.05256
	3/5	11.47110	1.82212	9.64898
	7/10	-43.39594	2.01375	45.40970
	4/5	215.87513	2.22554	213.64959
	9/10	-1002.80938	2.45960	1005.26898
	1	4732.66447	2.71828	4729.94618
1/20	0	1.00000	1.00000	0.00000
	1/10	1.11000	1.10517	0.00483
	1/5	1.34040	1.22140	0.11899
	3/10	4.15636	1.34986	2.80650
	2/5	67.54881	1.49183	66.05698
	1/2	1556.29600	1.64872	1554.64728
	3/5	36590.18000	1.82212	36588.35800
	7/10	861102.90000	2.01375	861100.88625
	4/5	20265860.00000	2.22554	20265857.77450
	9/10	476953600.00000	2.45960	476953597.54040
	1	11225020000.00000	2.71828	11225019997.28172
1/40	0	1.00000	1.00000	0.00000
	1/10	1.13628	1.10517	0.03110
	1/5	6.81247	1.22140	5.59107
	3/10	1108.15360	1.34986	1106.80374
	2/5	193992.03221	1.49183	193990.54038
	1/2	33834271.14782	1.64872	33834269.49910
	3/5	5917562236.00154	1.82212	5917562234.17942
	7/10	1034673659640.50269	2.01375	1034673659638.48894
	4/5	181044467862140.25000	2.22554	181044467862138.02446
	9/10	31676828316752360.00000	2.45960	31676828316752357.54040
	1	554682732167854100.00000 ⁸	2.71828	554682732167854097.28172

Question: 04

Consider a simple ecosystem of rabbits that has an infinite food supply and foxes that prey upon rabbits for their food. A mathematical model for this system consists of the following differential equations:

$$\begin{aligned}\frac{\partial r}{\partial t} &= 2r - \alpha r f, & r(0) &= r_0 \\ \frac{\partial f}{\partial t} &= -f + \alpha r f, & f(0) &= f_0\end{aligned}$$

where t is time, r is the number of rabbits, f is the number of foxes, and α is a positive constant. When α is zero, the two populations do not interact, and the rabbits do what rabbits do best, and the foxes die of starvation. This system has been studied for various values of α and initial conditions. For the purpose of this problem, we fix the value of α to be 0.01. To answer the questions in parts *a)* and *b)* below, use both second order Adams-Bashforth and Leapfrog methods to solve the differential equations. In each case use both the explicit Euler scheme and the second order Runge-Kutta scheme to start the integrations. Compare the results. In each case, what is the maximum time step for reasonable accuracy? What is maximum time step for reasonable accuracy?

a) Compute the solution with $r_0 = 300$ and $f_0 = 150$. You should observe that the behavior of the system is periodic with period very close to five time units. Plot the solution in the (f, r) plane. If the time step is too large, what happens to the periodicity of the solution?

b) Compute the solution with $r_0 = 15$ and $f_0 = 22$. What happens to the rabbit population? Integrate over several periods.

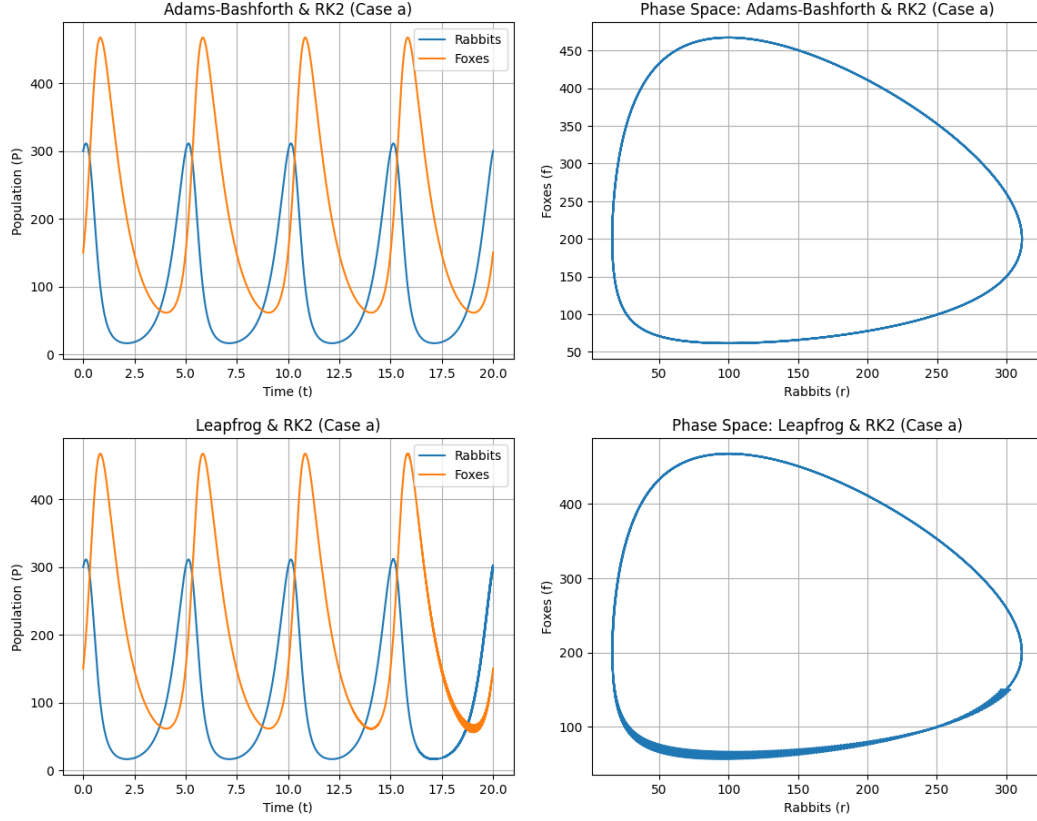
Solution:

The ecosystem dynamics between rabbits and foxes are modeled by the following differential equations:

$$\begin{aligned}\frac{\partial r}{\partial t} &= 2r - \alpha r f, & r(0) &= r_0 \\ \frac{\partial f}{\partial t} &= -f + \alpha r f, & f(0) &= f_0\end{aligned}$$

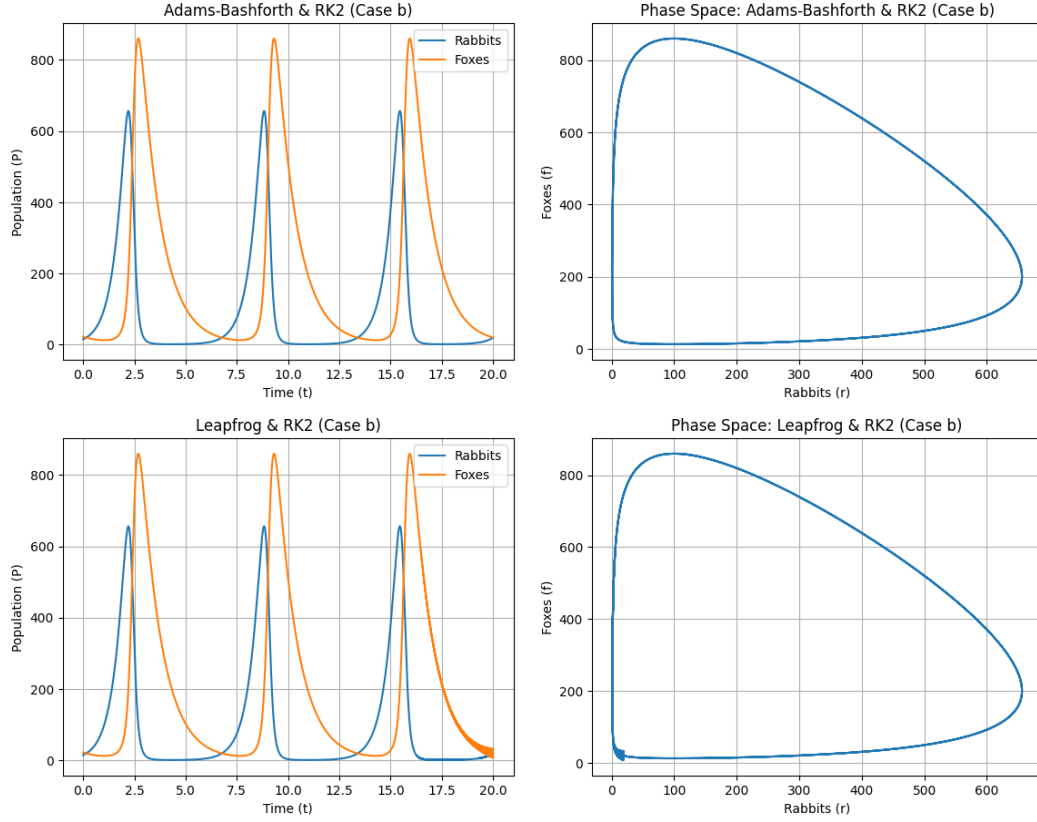
where t is time, r is the rabbit population, f is the fox population, and $\alpha = 0.01$ is a positive constant. The system is analyzed for two cases:

Case (a): $r_0 = 300$, $f_0 = 150$. Investigate the periodic behavior and plot the solution in the (f, r) plane.



Case (b): $r_0 = 15$, $f_0 = 22$. Study the behavior of the rabbit population over several periods.

The solution is obtained using the Adams-Bashforth and Leapfrog methods, initialized with both the Explicit Euler scheme and the second-order Runge-Kutta scheme (RK2).



Graphical Analysis:

Case (a): Initial Populations $r_0 = 300$, $f_0 = 150$

Adams-Bashforth & RK2

Time Evolution Graph:

- Rabbits ($r(t)$) and foxes ($f(t)$) exhibit periodic oscillations.
- The population cycles have a smooth periodic pattern, with a clear delay between the peaks of rabbit and fox populations.
- This indicates a predator-prey relationship where the rabbit population grows first, followed by a rise in the fox population.

Phase Space (Rabbit-Fox Plane):

- The trajectory forms a closed loop, representing a stable periodic behavior.
- The phase space graph shows the oscillations in population sizes are synchronized but offset in time.

Leapfrog & RK2

Time Evolution Graph:

- Similar periodic behavior is observed, but the oscillations may appear slightly more sensitive to initial conditions.
- The populations of rabbits and foxes alternate in peaks, demonstrating a typical predator-prey cycle.

Phase Space:

- The graph also forms a closed loop, reinforcing the stability of the periodic solution.

- Leapfrog tends to handle oscillatory systems better, ensuring accurate preservation of periodicity over longer simulations.

Case (b): Initial Populations $r_0 = 15$, $f_0 = 22$

Adams-Bashforth & RK2

Time Evolution Graph:

- The rabbit population ($r(t)$) declines initially, then stabilizes at a relatively lower oscillatory range.
- The fox population ($f(t)$) also stabilizes after a short transient period.
- This behavior suggests the ecosystem reaches a steady oscillatory dynamic, but with lower amplitude compared to Case (a).

Phase Space:

- A smaller, stable closed loop is observed in the phase space, indicating a steady oscillation between predator and prey populations.
- The system appears to converge faster to this steady state due to the lower initial populations.

Leapfrog & RK2

Time Evolution Graph:

- Leapfrog produces similar oscillatory results, but with potentially less numerical damping.
- The periodicity is preserved, with the same pattern of smaller oscillations compared to Case (a).

Phase Space:

- The phase plot shows a consistent and slightly smoother closed-loop trajectory.
- Leapfrog effectively maintains the integrity of the oscillatory dynamics over extended periods.

Question: 05

Solution

Single anharmonic system

The equation governing the single anharmonic system is:

$$\frac{d^2x}{dt^2} + \mu \frac{dx}{dt} + bx + cx^3 = k \cos(\omega t),$$

where the parameters are:

$$\mu = 1, \quad b = -10, \quad c = 100, \quad \omega = 3.5, \quad k = 1.55.$$

Conversion to First-Order System

Define:

$$y = \frac{dx}{dt}.$$

The system can be rewritten as:

$$\begin{aligned} \frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= -\mu y - bx - cx^3 + k \cos(\omega t). \end{aligned}$$

Numerical Integration

To analyze the system, numerical integration is performed using the Runge-Kutta method. The equations are solved over a long duration to allow transients to decay. Arbitrary initial conditions, such as:

$$x(0) = 1, \quad y(0) = 0,$$

Solution $x(t)$

A plot of $x(t)$ is created over time to demonstrate chaotic behavior. This irregular and aperiodic nature confirms the chaotic dynamics.

Poincaré Section

The system is invariant under:

$$x \rightarrow x, \quad y \rightarrow y, \quad t \rightarrow t + \frac{2\pi n}{\omega}.$$

The Poincaré section is constructed by plotting:

$$\left(x \left(\frac{2\pi n}{\omega} \right), y \left(\frac{2\pi n}{\omega} \right) \right),$$

at discrete intervals $t = n \frac{2\pi}{\omega}$.

Graphical Analysis:

- **Poincaré Section:** A scatter plot of sampled points in the phase space at intervals $t = n \cdot \frac{2\pi}{\omega}$, revealing the structure of the chaotic attractor.
- **Time-Series Plot:** $x(t)$, to demonstrate the irregular and aperiodic evolution of the state variable over time.

Poincaré Section

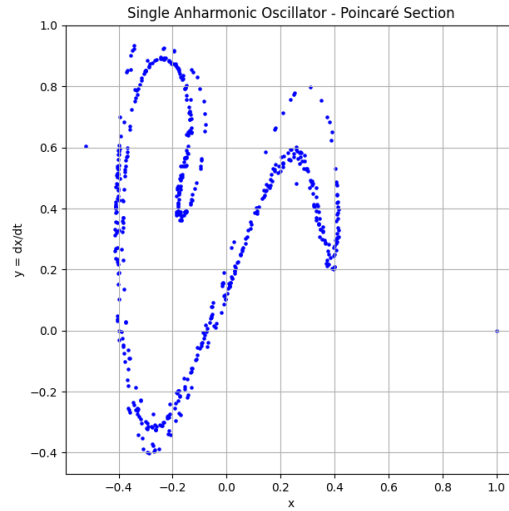


Figure 4: Poincaré section of the single anharmonic oscillator. Points sampled at $t = n \cdot \frac{2\pi}{\omega}$.

Description:

- The scatter plot in Figure 4 represents the Poincaré section.
- Points are sampled at regular intervals in the phase space, specifically at $t = n \cdot \frac{2\pi}{\omega}$.
- The plot reveals the structure of the chaotic attractor, showcasing a fractal-like distribution of points in the x - y phase space.

Time-Series Plot

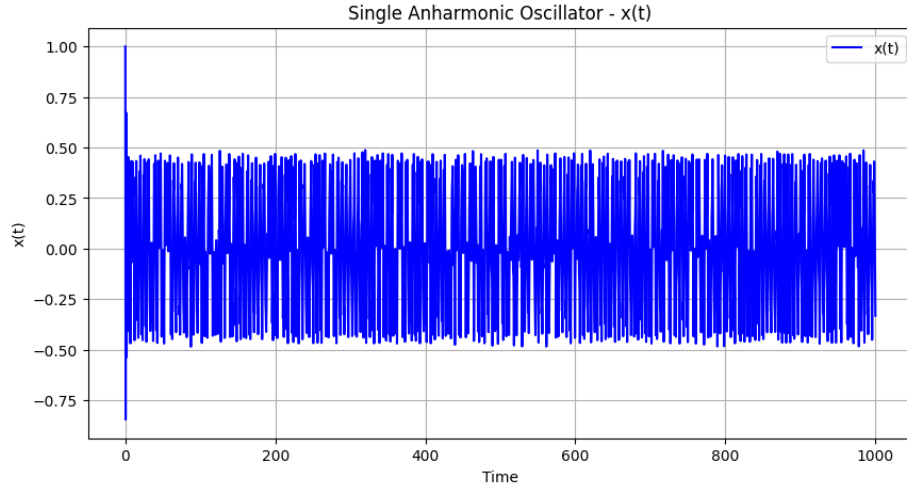


Figure 5: Time-series plot of $x(t)$ for the single anharmonic oscillator.

Description:

- Figure 5 displays the time-series plot of $x(t)$ over a long duration.
- The plot captures the evolution of x as the system progresses, highlighting irregular and aperiodic oscillations.
- The irregular behavior is characteristic of chaotic systems.

Coupled Anharmonic System

The coupled system is governed by:

$$\begin{aligned}\frac{d^2x}{dt^2} + \mu \frac{dx}{dt} + bx + cx^3 &= k \cos(\omega t) + g \frac{du}{dt}, \\ \frac{d^2u}{dt^2} + \mu \frac{du}{dt} + bu + cu^3 &= k \cos(\omega t) + g \frac{dx}{dt}.\end{aligned}$$

Conversion to First-Order System Define:

$$y = \frac{dx}{dt}, \quad v = \frac{du}{dt}.$$

The system becomes:

$$\begin{aligned}\frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= -\mu y - bx - cx^3 + k \cos(\omega t) + gv, \\ \frac{du}{dt} &= v, \\ \frac{dv}{dt} &= -\mu v - bu - cu^3 + k \cos(\omega t) + gy.\end{aligned}$$

Numerical Integration for Varying g

The system is solved numerically for different values of the coupling parameter g :

$$g = 0, 0.1, 0.2, 0.3, 0.4.$$

Solutions $x(t)$ and $u(t)$

Plots of $x(t)$ and $u(t)$ are created to observe the effect of varying g on the solutions.

Poincaré Sections

For each g , Poincaré sections are generated using:

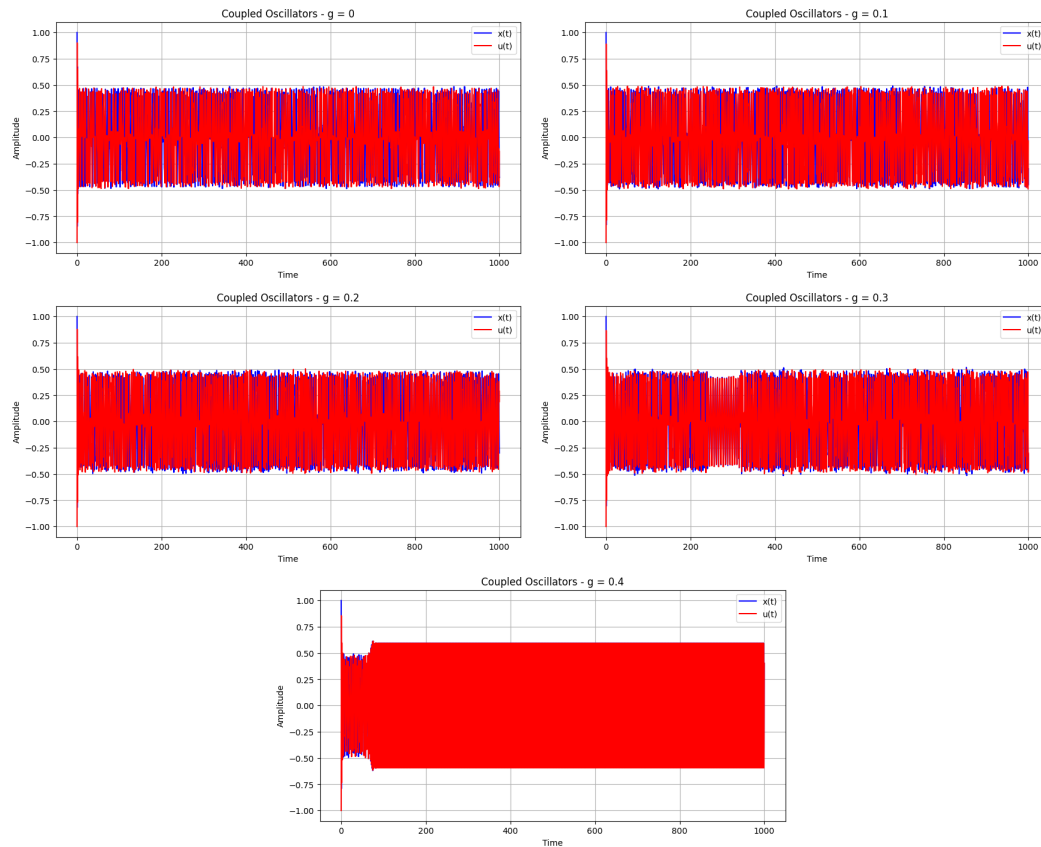
$$\begin{aligned}\left(x \left(\frac{2\pi n}{\omega} \right), y \left(\frac{2\pi n}{\omega} \right) \right), \\ \left(u \left(\frac{2\pi n}{\omega} \right), v \left(\frac{2\pi n}{\omega} \right) \right).\end{aligned}$$

Analysis of Coupling Effects

- For $g = 0$, the equations are decoupled, and the individual systems exhibit independent chaotic solutions.
- As g increases, synchronization effects and correlated dynamics emerge.

Time-Series Plots

Description: The time-series plots illustrate the evolution of $x(t)$ (blue) and $u(t)$ (red) over time for different values of the coupling parameter g .

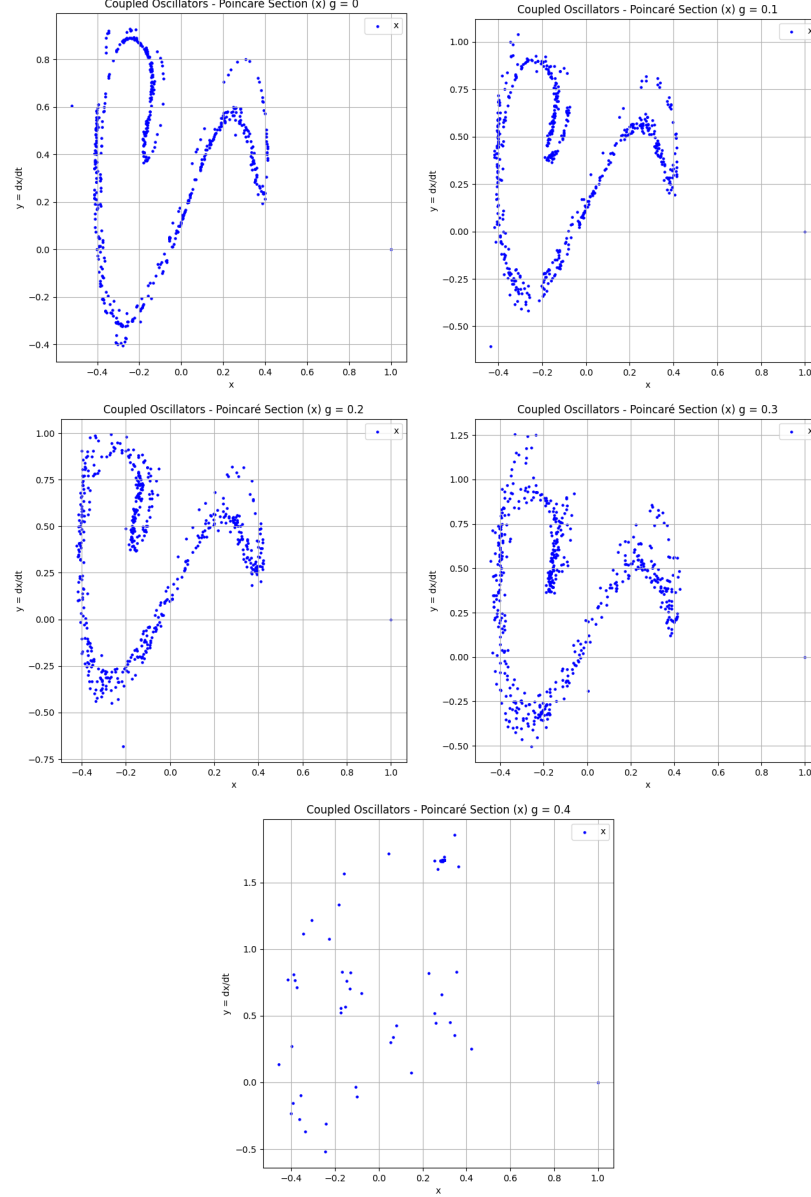


Observations:

- For $g = 0$, the oscillators are independent and exhibit chaotic behavior.
- As g increases, the dynamics of $x(t)$ and $u(t)$ become increasingly synchronized.

Poincaré Sections for x

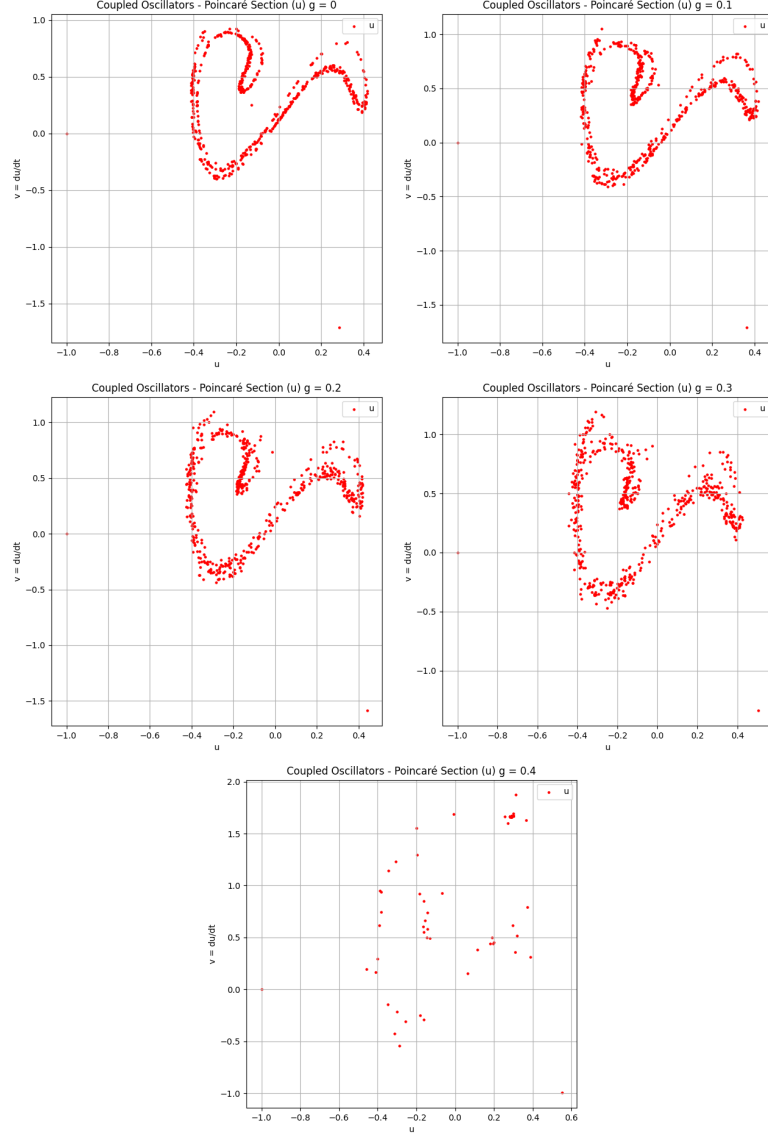
Description: The Poincaré sections display the sampled points in the phase space $(x, y = dx/dt)$ for different values of g .



Observations: The attractor exhibits a dense fractal-like structure at $g = 0$, characteristic of chaotic systems. As g increases, the structure evolves, reflecting the coupling's influence on x 's dynamics.

Poincaré Sections for u

Description: The Poincaré sections display the sampled points in the phase space $(u, v = du/dt)$ for different values of g .



Observations: Similar to x , the attractor for u is independent at $g = 0$ and shows chaotic behavior. The attractor's structure becomes more correlated with x as g increases, demonstrating the coupling's effect.